

Early and Late Fetal Growth Assessment & Well being

By

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Objective of fetal assessment:

The primary objective of antenatal assessment is to avoid fetal death. There is progressive decline in maternal and fetal mortality all over the world because of focusing on evaluation of maternal and fetal health and early detection of pregnancy risks.

Aims of assessment:

- To assess fetal growth and well being throughout pregnancy.
- To detect high risk factors that affect the growth of the fetus and multiparity.
- To show congenital abnormalities.
- Color Dopplar is a good tool for placental abnormalities and assess vessels
- Decision making for time and method of labour

Tools used in the assessment:

- Biophysical:
 - Ultrasonography (US):2D, 3D, 4D, Dopplar
 - CTG
 - Fetal Biophysical Profile (FBPP)
 - Clinical exam eg. Abdominal palpation, Symphysiofundal Height (SFH), Auscultation
 - Biochemical: α FP, bilirubin
 - Invasive procedures:
 - Amniocentesis
 - Chorionic Villus Sampling
 - Cordocentesis

For the purpose of better understanding, the pregnancy period divided into:

- Early: from zero to 20 completed weeks of gestation
- Late: from 20 weeks onward till delivery

Early pregnancy assessment

Ultrasonography (US):

In the early assessment, the following steps done:

- Confirmation of the pregnancy.
- Diagnosis of miscarriage, ectopic pregnancy
- Early embryonal growth
- Detect early congenital abnormalities.

Diagnosis and confirmation of fetal viability by US in the early weeks is done with different findings in each week period:

<i>Weeks</i>	<i>Findings</i>
4-5	Gestational sac
5	Yolk sac
5-6	Embryo seen & measured
6	Heart Beat

Transvaginal ultrasound is useful in early pregnancy especially in the diagnosis of early pregnancy failure and in fatty abdominal wall, for examining the cervix later in pregnancy and for identifying the lower edge of the placenta. Early pregnancy failure like Missed miscarriage , Blighted Ovum , Ectopic pregnancy.

All pregnant women should be offered routine scheduled scans at between 10-14 weeks and 18 - 21 weeks gestation. Gestational age & Fetal maturity evaluation by US is done by using the following parameters:

- At 5-10 weeks: gestational sac.
- At 8-14 weeks: the crown-rump length (CRL)
- At 14-20 weeks: Femur Length (FL), Head circumference (HC), Abdominal circumference (AC). HC is the best determinant of the Gestational Age after W14

Major fetal structural abnormalities occur in 2–3 % of pregnancies. US examination detect fetal anomalies around or before 20 weeks gestation like:

- Neural Tube Defects eg. spina bifida, hydrocephalus, meningocele
- Skeletal abnormalities eg. achondroplasia
- Abdominal wall defects eg. exomphalos, gastroschisis
- Cleft lip/palate
- Congenital cardiac abnormalities.

Important findings in the early US examination of pregnancy which may clue the presence of fetal abnormalities are:

- If CRL < gestational age , this will increase the risk of chromosomal anomalies (trisomy or triploidy).

- Increase nuchal translucency (soft tissue marker) at 10-14 weeks is associated with many chromosomal abnormalities (trisomy like Down Syndrome, monosomy, triploidy).

Biochemical:

Maternal serum &/or amniotic level measurements of chemicals is a tool used for early detection of fetal diseases. Parameters assessed like ; α FP, HCG, Unconjugated estradiol(UE), Acetylcholine esterase (AChE), bilirubin...etc. Raised or declined levels of these proteins are nominators of possible pathology in the fetus that need further evaluation

Alfa Fetoprotien (α FP):

It is synthesized from yolk sac and fetal liver. It reaches peak concentration in amniotic fluid or fetal serum in 13 weeks gestation and peak concentration in maternal serum in 32 weeks gestation. Detection of raised or declined level is an indicator of possible fetal pathology.

Raised α FP seen in

1. Wrong gestational age
2. Open neural tube defects (NTDs)
3. Multiple pregnancy
4. IUFD
5. Renal anomalie

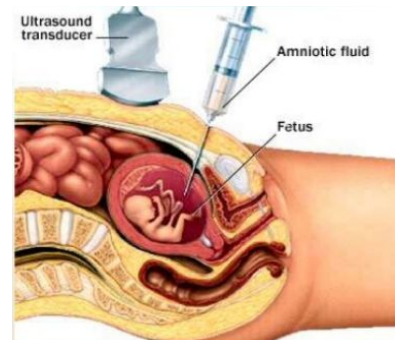
Decreased α FP seen in

Down Syndrome

Invasive procedures:

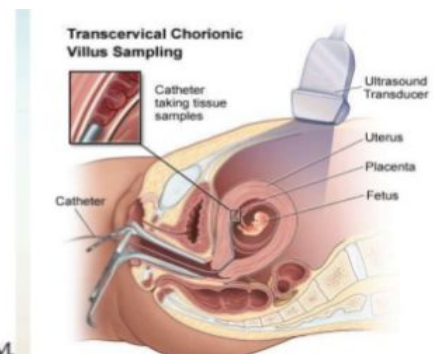
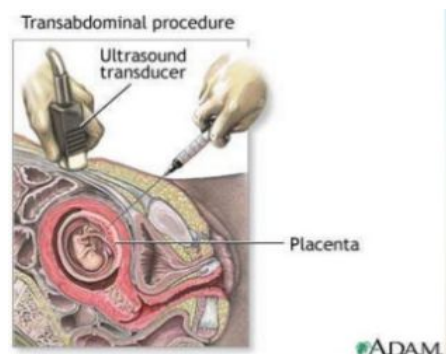
Amniocentesis:

It is aspiration of amniotic fluid from gravid uterus which is typically scheduled between the 14-16 weeks of pregnancy. US is done to determine the position of the fetus , and then the pocket of amniotic fluid and the placenta found and needle aspiration of amniotic fluid is taken for biochemical analysis mainly for α FP, bilirubin or Chromosomal analysis.



Chorionic villus sampling CVS

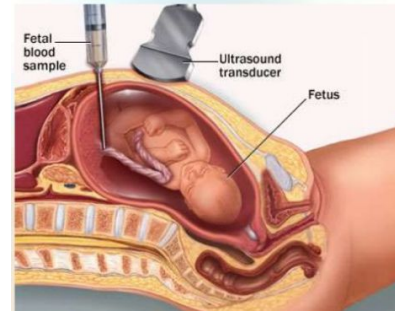
It is performed for prenatal diagnosis of genetic disorders. It is carried out transcervically between 10-12 weeks and transabdominally from 10 weeks to term. A few villi are collected from the chorion frondosum under US guidance. It provides earlier diagnosis than amniocentesis but complications like fetal loss(1-2%) or vaginal bleeding are higher.



Fetal blood sampling (cordocentesis)

In this technique a sample of fetal blood taken during pregnancy in order to screen for chromosomal abnormalities, hemoglobinopathies and cytology. It is performed under local anaesthesia usually after 18 weeks gestation.

Risks: Its invasive procedure may lead to abortion, preterm labour and IUFD, due to bleeding, cord haematoma, infection or preterm rupture of membranes.



Multiple pregnancy:

US is best method of diagnosis. It is used in identifying the presence of more than one fetus and determine the chorionicity of the gestation.

- Mnochorionic pregnancy associated with more complications than Dichorionic like higher perinatal mortality
- Presence of septa in monochorionic is formed by two amniotic membrane while of dichorionic are both two amnion and two chorion

Late pregnancy assessment

Monitoring of fetal movement (FM) :

It is the simplest, cheapest, and most continuous test of fetal wellbeing but it depends on the mother's Perception, and her concerns should always be taken seriously.

- Primigravida perceive FM → 18 weeks onward
- Multigravida perceive FM → 16 weeks onward

In 55% of IUFD cases are preceded by ↓ FM. A healthy fetus moves with a degree of consistency, or at least 10 times a day. Loss of fetal movements is commonly followed by disappearance of FHR within next 24 hours. Although routine FM counting did not ↓ perinatal mortality, but still mothers encouraged to do so as early alarm fetal possible distress.

Clinical examination:

Symphysio Fundal Height (SFH):

It is the best method to indicate whether the baby is small or large for dates compared to assumed gestational age. SFH should be plotted on a growth chart in the notes which improve prediction of SGA neonate.

- 1 week = 1 cm height from pubic bone

Abdominal palpation:

It is feeling of fetal parts, lie and engaging part near the time of Labour . Bimanual examination is also a part of palpation and any uterine abnormality also can be felt like fibroid.

Auscultation:

Using equipments like Pinard and Sonic aid.

Typically you can hear the baby's heart beat from about 18-20 weeks onward, depending on maternal and fetal factors (weight of mother, the position of your baby, the location of the placenta, etc.).

Cardiotocograph CTG:

Fetal cardiac behaviour is regulated through the autonomic nervous system and by vasomotor, chemoceptor and baroreceptor mechanisms. Pathological events, such as fetal hypoxia, modify these signals and hence cardiac response including variation in heart rate patterns, which can be detected and recorded in the CTG. Features which are reported from a CTG to define normality and identify abnormality and potential concern for fetal well-being include the:

- Baseline rate
- Baseline variability
- Accelerations
- Decelerations

Interpretation of the CTG should always be made in the context of clinical situation. Two tests performed

- Non stress test: interpreted as;
 1. Reassuring
 2. Non reassuring
- Contraction stress test:
 1. Negative
 2. Positive

Ultrasonography (US):

In the late assessment, US assessment of growth and fetal wellbeing includes:

- Fetal growth pattern and weight estimate from biometric measurements
- Measurement of liquor volume
- Fetal breathing, movements and tone, and any gross abnormality seen of fetal parts
- Doppler assessment of Umbilical Artery.

US examination used in Biometric measurements for assessing fetal growth:

- Fetal abdominal circumference (AC)
- Biparietal diameter (BPD)
- Head circumference (HC)
- Femur length (FL)

Combining the above parameters results in Estimated fetal weight (EFW).

Fetal growth and wellbeing assessment according to EFW will be as following:

- W 24: 500 g W28: 1 Kg W40: 3.5 Kg

A discrepancy of 15% may exist between EFW and actual fetal weight. The AC is the most sensitive predictor of fetal weight. Individual biometric measurements are less predictive for IUGR than the overall fetal growth velocity.

The main source of amniotic fluid is fetal urine after W20. Any disruption to fetal urine production, passage, or swallowing can alter the volume. The volume according to different gestational age:

- Objective measurements include

- | Oligo | Poly |
|------------------------|------------------------|
| <2 cm | >25 cm |
| < 5 th Cent | >95 th Cent |

US for localizing placenta and exclude Placenta previa is now become a part of routine clinical practice to avoid APH

- At 20 weeks → 5% of pregnancy has Low lying placenta (LLP)
- Most of LLP migrate upward as lower uterine segment stretches from late 2nd trimester and 3rd trimester
- Only 5% by the time of delivery ended up to have Placenta previa and so risk of APH

- Cervical length
- Confirmation of IUD
- Confirmation of fetal presenting part in uncertain cases
- Diagnosis of other pathologies like fibroid, ovarian cyst during pregnancy

DUS allows the assessment of the blood flow within fetal and placental vessels and provides indirect assessment of fetal and placental condition. It is depending on difference in velocity and direction of RBC flow toward or away from transducer, and displayed in graphic form, with the velocity plotted against time. Analysis of downstream resistance in the peripheral vessels done instead of depending on flow velocity alone.

DUS demonstrates forward flow of blood throughout the systole and diastolic phases of cardiac output.

- UA Flow changes conversely with X 1/ Resistance Placenta
- | | | |
|--------|---|--|
| Normal | ↓ | → Normal & Healthy |
| ↓ | ↑ | → Infarcted Placenta
IUFD, Fetal Distress |

Fetal Growth & wellbeing Assessment

Fetal vessels:

Decrease fetal blood oxygen level caused Centralization of blood flow to the vital organs such as Brain, Heart, Adrenals. In DUS the findings will be:

- Hypoxia= $\uparrow$ Middle Cerebral Artery (MCA) + $\downarrow$ Aorta+ Vasoconstriction
- Acidemia & Heart Failure= $\uparrow$ MCA + Absent Aorta+ Pulsatile Ductus venosus and Inferior Vena Cava
- Impending Fetal Death= Absent Ductus venosus which indicate Immediate Delivery
- MCA Systolic Velocity $\uparrow$ in fetal anaemia.

Uterine Artery -- UtA:

First trimester & early 2nd trimester DUS of UtA may predict Preg. Outcome

- Diastolic waveform Notch means $\uparrow$ resistance of UtA and so increasing resistance of Uteroplacental bed
- $\uparrow$ resistance of Uteroplacental bed means high possibility of pre-eclampsia, IUGR and placental abruption $\rightarrow$ more follow up needed
- Bilateral UtA Notching at 20-24 WK $\rightarrow$ 60-70% chance of above complications specially PIH

Biophysical profile (BPP):

It's a 30 min US scan for Fetal behavior (fetal breathing, gross body movements, flexor tone), Amniotic fluid (AF) volume and CTG (Accelerations). The parameters are:

F breathing movement (Score 0 or 2)

F gross body movement (Score 0 or 2)

F Tone (Score 0 or 2)

AF (Score 0 or 2)

CTG (reactive heart) (Score 0 or 2)

Interpretation:

- 0, 2 or 4 is considered abnormal
- 8 or 10 normal
- 6 equivocal need to be repeated within Hours

Disadvantages

- Time consuming (30 min) in High risk Pregnancy
- The may be sleep and so the score will be low
- When score is low, means fetus is already in critical situation

So studies concluded that there is insufficient support the use of BPP as a test of fetal well-being in **high-risk pregnancies.**

MRI

Use of MRI during 1st trimester of pregnancy should be restricted to maternal indications for which the information is considered clinically vital. Fetal MRI is safe at 3.0 tesla or less during 2nd & 3rd trimesters.

Gadolinium contrast (classified as a category C drug) may be used in pregnant women when the benefits outweigh the potential risks. It is safe to continue breastfeeding after receiving a gadolinium contrast agent.

GOOD LUCK